Vishnu Varthan VG

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About Me

Electrical and Electronics Engineering student focused on hardware manufacturing, embedded systems, and semiconductor engineering, with hands-on experience in Arduino-based projects, motor control, and industrial electrical systems. Eager to apply technical knowledge to real-world engineering challenges and develop expertise in hardware design and automation.

Education

Chennnai Institute of Technology Expected May 2029
Bachelor of Electrical and Electronics (CGPA: 8.30 / 10.00) *Chennai, Tamilnadu*

- **Relevant Coursework:** Digital Electronics, Microprocessors and Microcontrollers, Electrical Measurements and Instrumentation, Control Systems Engineering, Power Electronics

Experience

Auto Print Machinery May 11 2026– May 19 2026
Assistant Electrical Engineer *Coimbatore, Tamilnadu*

- Assisted in the inspection and testing of industrial electrical components, including MCBs, overload relays, contactors, and variable frequency drives (VFDs).
- Observed and supported three-phase motor control wiring, including connections between contactors, MCBs, overload relays, and motors, to understand industrial switching and protection systems.
- Gained practical exposure to industrial electrical panels, SMPS, PLCs, HMIs, EMI filters, and line chokes, developing an understanding of electrical system integration and troubleshooting.

Projects

Flame rover | *IR Flame Sensors, Motor Driver, Relay Module*

- Developed an Arduino-based flame detection rover using three IR flame sensors to detect fire sources from the left, front, and right directions.
- Implemented sensor-based motor control using a motor driver to enable directional movement toward detected flame sources.
- Integrated a relay-controlled water pump to support fire suppression, combining flame detection, rover movement, and pump activation in an automated prototype

PID-Based DC Motor Speed Control | *Arduino Uno, Embedded C/C++, PID Control, PWM, L298N, Hall-Effect RPM Sensor*

- Successfully implemented a closed-loop DC motor speed control system using a Proportional-Integral-Derivative (PID) algorithm to maintain the desired motor speed.
- Integrated a potentiometer for target RPM selection, a Hall-effect sensor for real-time speed feedback, and an L298N motor driver to dynamically adjust motor speed through PWM control.

ESP32-Based Smart Safe Locker | *ESP32, Embedded C/C++, OLED, GPIO, Security systems*

- Developed an ESP32-based electronic safe locker featuring potentiometer-based passcode entry, OLED status display, and red/green LED indicators for access feedback.
- Implemented a timed security lockout that disables further access attempts for 30 seconds after three consecutive incorrect passcode attempts.

VisionShield | *Python, Machine Learning, Computer Vision, Dataset Validation, Model Integrity, Cryptographic Auditing*

- Designed VisionShield, an offline, model-agnostic computer vision assurance framework to identify potential dataset integrity issues, including data poisoning, mislabelling, duplicate samples, and anomalous data.
- Proposed model integrity checks to detect potential backdoors, unauthorized model substitution, distribution shifts, and inference-time tampering across computer vision workflows.
- Developed a risk assessment approach incorporating confidence scoring, review and quarantine recommendations, and cryptographic audit trails to support traceable and trustworthy model evaluation.

MOSFET Thermal Analyzer | *Python, Power Electronics, Semiconductor Devices, Thermal Analysis*

- Developed a Python-based simulation tool to analyze MOSFET power losses and thermal behavior under different operating conditions.
- Evaluated semiconductor conduction and switching losses and estimated device temperature rise to support thermal reliability analysis and component selection.

ML-driven adaptive noise cancellation | *ML, Digital Signal Processing (DSP), ANC, FIR Filtering, Embedded Systems*

- Designed an ML-driven adaptive noise cancellation system to classify stationary, non-stationary, and impulsive noise for improved noise suppression while preserving speech intelligibility.
- Developed a hybrid noise suppression approach combining ML-based noise classification with fast DSP-based impulse detection and FIR filtering for responsive noise reduction.

Technical Skills

Programming Languages: C, C++, Verilog HDL, Python
Microcontrollers & Hardware: Arduino Uno/Nano, ESP32, DC Motor Drivers (L298N), IR Flame Sensors, Relay Modules, ADC, PWM, I2C, UART
Electrical Engineering: Electrical Machines, Circuit Assembly, Motor Control, Electrical Measurements, Transformer Connections, Power Electronics Fundamentals
Industrial Automation: MCB, OLR, Contactors, VFD, SMPS, PLC and HMI Fundamentals, EMI Filters, Line Chokes
Tools & Platforms: Arduino IDE, Serial Monitor, GitHub, Antigravity
Technical Concepts: Embedded Systems, Open-Loop Control, Closed-Loop Control, Digital Electronics, State-Space Analysis, Digital Signal Processing Fundamentals, Machine Learning Fundamentals, Computer Vision Assurance

Achievements

- Secured 2nd place in the VISION X Hackathon.
- Finished as Runner-up in the Y'HACK Hackathon.